

Global Inference over Boundary-Head Candidates: A Research Programme

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Programme map, revision of 2026-08-28. States the unifying thesis, the three papers it decomposes into, what is established, and what is still open. Numbers quoted here are summaries; each one's primary record is the working paper cited beside it.

1. The thesis

GLiNER2's **boundary head** emits candidate scores per chunk — span boundaries, mention and pair logits, and contextual candidate states — and a **greedy per-chunk decode** then selects a record set from them. That decode is adequate for sparse, single-document extraction. The programme's claim is that it is not adequate for anything denser:

Dense document-level and beyond-document extraction require global structured inference over the boundary candidate scores, not the greedy per-chunk decode.

The claim has two instantiations. They are the *same* top-K hypothesis inference at different scopes, over one shared `candidate_scores` → `JointProblem` contract:

scope	mechanism	what the beam ranges over	target task
within a document (combinatorial)	global decode under typed constraints	top-K constraint-consistent assignments	events (RAMS) and relations (Re-DocRED)
across documents (temporal)	EKF/MHT tracker — Kalman bank plus pruning	top-K hypotheses over time	evolving events in a document stream

Within a document, the two faces are one mechanism: in the beam an event is a *trigger node plus role edges* and a relation is a *plain edge*, so a single typed-constraint decode covers both. A win on only one face is a weaker but still reportable result, and the honest negative — that global decoding helps relations and not events, or the reverse — localises where greedy per-query decoding actually costs you.

Across documents the row is two questions, not one: **track** an evolving quantity, and **diarise** observations into the right stream. The distinction matters because the two halves fail independently, and the second can silently destroy the first.

The boundary head is the shared substrate for both. It removed the span architecture's 19-instances-per-type cap, which had made either mechanism impossible to express.

2. What is established

The span head, not the encoder, was the bottleneck. Holding fresh heads fixed and swapping the encoder (mmBERT ↔ DeBERTa-v3) moves RAMS argument F1 barely at all (0.050 vs 0.042). Holding the encoder fixed and swapping fresh → IE-pretrained heads moves it roughly elevenfold (0.042 → 0.462). The deficit is head initialisation, not context window. (*Paper 0*, §10.5.)

Head warming has a measurable data threshold. Warming mmBERT's heads on a structure/argument corpus before RAMS fine-tuning does nothing at 10K, lifts arguments about $2.3\times$ at 40K, and keeps climbing to 100K with no plateau. The knee is between 10K and 40K. (*Paper 0*, §10.7.)

Windowing, not the global decoder, recovers document-level arguments on a short-context model: 0.086 → 0.144 argument-strict F1 from matching the eval window to the trained window, against which the OneIE-style beam is neutral (0.144 → 0.137). Reported as a negative result for the decoder. (*Paper 0*, §10.2.)

Replay protects a warm start, and record extraction transfers without record supervision. Warm-starting the 137K joint base on real-plus-synthetic data with 30% exact replay held the original structure capability (0.1119 → 0.1060 on the base's own test set) while nearly tripling on the new distribution (0.0755 → 0.2179). The new corpora supplied *no* record-head supervision at all, so the gain is transfer from the span representations the record head reads its field fillers out of. (*JOINT_IE_SCALING.md* §0b–0c.)

An uncalibrated operating point cost more than any training intervention. (2026-08-27.) The stage-0 relevance gate ran its whole life at threshold 0.5. Its softmax is saturated and it needs 0.998 to sit at the stated recall bar; choosing the threshold on validation and scoring the blind test once moves overall accuracy 0.719 → 0.847 and the worst class 0.444 → 0.903, fixing 108 rows against 39 broken (exact McNemar $p = 1.1 \times 10^{-8}$). Two GPU fine-tuning runs and a four-way auxiliary label had been spent on that class; at each model's own validation threshold the rebuilt gate and its predecessor are indistinguishable (18/18, $p = 1.0000$). This is the §5 rule "quote a curve only with the operating point it was read at" recurring as a several-hundred-dollar lesson rather than a misdiagnosis. (*GATES.md*; *EKF_MHT_DESIGN.md* §5.1.)

Single-run variance on these metrics is ± 0.02 or worse. A control re-run of a published RAMS recipe scored +0.023 above it from the re-run alone. Any curve claim needs at least two seeds per point before it is quoted, and several earlier readings of this programme's own curves did not meet that bar.

3. What is not established

The central question is now asked and answered — and the specified design was the wrong answer. (Updated 2026-08-25.) The design specified association as gate → Hungarian assignment → top-K hypotheses → track birth/death, and for a long time what shipped was hard assignment on an observable string key: no hypothesis enumeration, no deferred decision, no track birth or death. Every real-event failure recorded below is a failure of that placeholder.

What replaces it is **not** the specified design. A **global Viterbi decode over three states — own place, aggregate, reject** — improves every event we have at one setting: Helene 29.3 → 20.7 (−29.4%), Türkiye–

Syria 11,581.5 → 10,695.5 (−7.6%), Aegean 2020 74.4 → 15.7 (−78.8%). Two properties carry it, and a hypothesis tree is neither: the decision must be **global**, because a greedy rule commits per observation and one large figure admitted early poisons a stream's scale for everything after; and it must be able to **reject**, because assignment headroom is measured at **zero** and the entire residual is the null hypothesis. Hungarian assignment optimises the half that is worth nothing. (*EKF_MHT_DESIGN.md* §7.7.)

On real news the tracker loses to a trivial baseline. On the Türkiye–Syria 2023 earthquake, pre-registered, `est_last_value` beats the EKF (0.208 vs 0.136), a 1999 death toll quoted in an article's history section is tracked as a 2023 figure, and one of the two affected countries is never recovered at all. Attribution, not filtering and not extraction, is the bottleneck. (*EKF_MHT_DESIGN.md* §4.)

The greedy-vs-beam comparison — the actual question of the combinatorial arm — has not been run. Every number produced so far is the greedy arm.

Structure supervision is not reaching the record head from most corpora. The `cc_news` and synthetic converters emit structures without the metadata the training path needs, so records that appear to supply structure supervision supply none. This is a data-design decision rather than a defect to patch, and it gates any future arm claiming structure capability from those corpora.

4. The three papers

Paper 0 — Foundation. *Schema-Driven Information Extraction Beyond the Sentence*. GLiNER2 plus the boundary head plus head-initialisation and multi-corpus training: the substrate the other two build on. Complete; no new experiments required.

Paper 1 — Real-time events (temporal). The EKF/MHT line: censored measurement model, learned gate, text-to-observation normalisation, held-out synthetic validation, then a genuinely blind real event. Most mature on the tracking half. Blocked on the diarisation half, which is the honest statement of where it stands. Design and results: *EKF_MHT_DESIGN.md* .

Paper 2 — Traditional events (combinatorial). Global decode wired to the boundary head, measured on both RAMS (events) and Re-DocRED (relations), then structured joint training. The base-volume × architecture curve is complete on repaired data; the decode arm is not. Design and results: *JOINT_IE_SCALING.md* .

Papers 1 and 2 share the framing in §1 and cross-cite; a later extended version may merge them into the single global-inference statement.

5. Method rules adopted after being learned the hard way

Verify splits before every run. A per-row random draw scattered copies of the same document across train, validation and test in 45 corpora. It is silent — nothing crashes and no metric looks wrong. Within-split overlap is now gated automatically before training, and Paper 0 §7.1 reports the audit and its effect on that paper's results rather than quietly restating the numbers.

Never compare across test sets. Two of this programme's retracted findings were cross-row comparisons on differently-composed blind tests. Support counts must match, or the comparison is not one.

Always report the trivial baseline beside the model. A pre-registered EKF prediction was scored a success by a range-normalised metric that was blind to contamination landing mid-range; `est_last_value` alongside it would have caught this immediately.

Quote a curve only with the operating point it was read at. A head whose decode threshold is never calibrated can read exactly zero while working correctly — which is how one head in this programme was misdiagnosed as broken across five measurements. It recurred in 2026-08-27 on the stage-0 gate, where the default threshold cost more accuracy than two fine-tuning runs recovered, and where a model comparison reported as significant ($p = 0.0091$) vanished entirely (18/18, $p = 1.0000$) once each model was read at its own threshold. **A comparison between two models at a shared arbitrary threshold measures the gap between two operating points, not between two models.**

A stored verdict is valid only for the model it was measured on. An ablation concluding that translation does not repair a multilingual gate was correct for the model it ran on — which reads the language and over-admits — and exactly wrong for a later model that cannot read the language at all, where translation moved AUC $0.4733 \rightarrow 0.8359$. Re-run a cited verdict before applying it to a model that was not in it.

Graduation rule. A finding lives in its working paper until verified held-out, then graduates into the target paper. The working papers stay as the design and decision record; the papers carry only verified, reproducible claims — including the honest negatives that scope the contributions.

6. Source documents

The working papers are the primary record; this map is a summary of them.

document	role
PAPER_0_FOUNDATION.md	Paper 0 draft — substrate, head-init finding, data-integrity audit
JOINT_IE_SCALING.md	Paper 2 — joint decoding, scaling curve, replay, the three silent defects
JOINT_IE_DESIGN_RECORD.md	Paper 2's build record — decisions, wiring map, cost model, deferred Phase B
EKF_MHT_DESIGN.md	Paper 1 — the filter, the real-event defeat, and the association half rebuilt as a global decode (§7.7)
EKF_MHT_BUILD_RECORD.md	Paper 1's build record — attachment points, generator spec, blind-test protocol
GATES.md	every gate in the pipeline, its type, and whether it is on
THIRD_EVENT_AEGEAN2020.md	the third event: pre-registration, and the prediction it falsified
BOUNDARY_ARCHITECTURE.md	how the boundary head works end to end, per task
COUNTING_LAYER.md	why the span 19-instance cap is a dead end
BOUNDARY_DECODE_AND_EKF.md	verified decode map and where global inference attaches
KALMAN_BEAM_SEARCH_EXPLORATION.md	the beam ↔ filter origin analysis
PROJECT_JOURNAL.md	chronological record of decisions, including those later overturned
TODO.md	open defects and next tests, with the evidence for each
PIPELINES.md	as-built vs as-designed maps, and the divergence table with M1–M5 status